

AI VPP Property Intelligence Report

Executive Summary

This property shows strong solar feasibility potential due to its estimated rooftop area of 13805 sqm and average solar irradiance of 6.0 Peak Sun Hours. The estimated solar system can generate approximately 5829 kWh/day, making the property suitable for commercial solar deployment and future VPP participation.

Property Analysis

Property Type: Mall

Roof Area: 13804.57 sqm

Building Type: stadium

Building Name: Sree Kanteerava Outdoor Stadium

Building Levels: 1

Estimated Daily Consumption: 8282.74 kWh

Consumption Methodology: Estimated using commercial energy intensity benchmarks based on property type and built-up area.

Solar Feasibility Analysis

Peak Sun Hours: 6.0

Usable Rooftop Area: 11043.66 sqm

Estimated Solar Capacity: 1214.4 kW

Estimated Panel Count: 2208

Daily Energy Generation: 5829.12 kWh

Annual Energy Generation: 2127628.8 kWh

Homes Powered Equivalent: 485

EV Charging Equivalent: 97

Solar Insights: The rooftop appears suitable for medium-to-large scale commercial solar deployment.

Battery Storage Analysis

Battery Capacity: 1.75 MWh

Battery Size: 1748.74 kWh

Battery Insights: Recommended battery sizing supports energy shifting, backup support, and VPP participation.

Financial Analysis

Annual Savings: \$255315.46

Estimated ROI: 3.8 years

Financial Insights: Estimated savings are based on projected solar generation offsetting grid electricity.

Environmental Impact

Carbon Savings: 1489.34 tons/year

Equivalent Trees Planted: 67020

Environmental Insights: The project can significantly reduce annual carbon emissions.

Virtual Power Plant (VPP) Analysis

VPP Score: 67 / 100

Readiness Level: Moderate

Summary: The property has moderate VPP potential with opportunities for optimization.

Supported Grid Services:

Peak Shaving, Demand Response, Backup Power, Grid Export Support

Key Strengths:

Strong commercial solar generation potential.

Excellent regional solar irradiance.

Commercial property profile aligns well with distributed energy deployment.

Recommendations:

This report was automatically generated using the AI VPP Property Intelligence Engine.